# Numerical Investigation of Split and Recombination (SAR) Microchannel Mixers

ME 224, Fundamentals of Microscale Flows

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## Abstract

Efficient mixing in microchannels is challenging due to laminar flow conditions and diffusion-dominated transport at low Reynolds numbers. In this study, the mixing performance of three representative types of Y-shaped split and recombination three-dimensional passive micromixers of different split-and-recombine (SAR) micromixer geometries is numerically investigated. The incompressible Navier–Stokes equations are solved to obtain the velocity field, and the convection–diffusion equation is used to determine species concentration distribution. The mixing index is evaluated at the outlet cross-section for Reynolds numbers ranging from 1 to 3. The mixing indexes for each micro-mixer design are reviewed in ANSYS 2025R2. All proposed designs combining index line graphs and images clearly show that mixing is much improved compared to the straight mixer (i.e., Geometry 1). Results indicate that mixing initially decreases with increasing Reynolds number due to reduced residence time and limited diffusion. However, beyond a critical Reynolds number, enhanced convective transport and transverse flow significantly improve mixing performance. Among the studied configurations, Geometry 3 demonstrates superior mixing efficiency due to increased interfacial area and more substantial recirculation effects. The study highlights the importance of geometric modifications in enhancing micromixing under laminar flow conditions.

# Contents

# 1. Introduction

Microfluidic systems facilitate the manipulation of fluids within microscale channels and are widely used in applications such as lab-on-a-chip devices, biomedical diagnostics, and microreactors. A fundamental requirement for the efficacy of these systems is the efficient mixing of fluid streams. However, the small characteristic dimensions of microchannels typically result in laminar flow ($Re < 100$), where turbulence is absent and mixing predominantly relies on molecular diffusion.

In laminar flow, fluids traverse the channels in parallel layers, with minimal transverse motion, leading to limited mixing primarily governed by diffusion across fluid interfaces. To enhance mixing without introducing additional complexity to the system, passive micromixers employ geometric modifications to improve fluid interactions. Among various passive designs, split-and-recombine (SAR) micromixers are particularly effective; they enhance mixing efficiency by repeatedly dividing and recombining fluid streams, thereby increasing interfacial area and promoting transverse flow and chaotic advection.

The mixing performance of SAR micromixers is significantly influenced by channel geometry, which affects both fluid residence time and pressure drop. While geometric modifications may enhance mixing efficiency, they can also increase hydraulic losses. Therefore, a systematic evaluation of different SAR geometries is critical to identify designs that achieve adequate mixing while maintaining acceptable pressure penalties.

This study presents a comprehensive numerical investigation of split-and-recombine microchannel mixers across three distinct geometrical configurations: a baseline straight channel, a circular SAR channel, and a rhombus SAR channel. Simulations are conducted under steady, laminar flow conditions at varying Reynolds numbers, and the mixing performance is assessed using the mixing index to elucidate the influence of geometry on micromixing behaviour.

## 2. Literature Review

Efficient micromixing in microfluidic systems poses significant challenges because turbulence is absent in microchannels, where mixing is primarily governed by molecular diffusion. To address this limitation, a variety of passive micromixer designs have been devised, enhancing mixing capabilities through geometric modifications. Notably, split-and-recombine (SAR) micromixers optimize mixing by repeatedly dividing and recombining fluid streams, thus increasing the interfacial area and minimising the diffusion length.

Mahmud (2021) conducted a numerical investigation of a Y-U split-and-recombine micromixer at Reynolds numbers ranging from 1 to 50, reporting mixing efficiencies of up to 90% under diffusion-dominated conditions. The study found that increasing connector height improved mixing by elongating the flow path; however, this also increased pressure drop, underscoring the inherent trade-off between mixing efficiency and hydraulic performance.

In a complementary study, Shah et al. (2019) experimentally and numerically investigated the performance of Y-shaped SAR micromixers with straight, circular, rhombus, and hybrid geometries over a Reynolds number range of 0.5 to 100. The results indicated that straight channels exhibited suboptimal mixing performance (approximately 15%), while the circular and rhombus geometries significantly enhanced mixing efficiency, achieving values close to 99% at elevated Reynolds numbers. The observed enhancement in mixing performance was attributed to the generation of secondary flows and chaotic advection resulting from channel curvature and the repeated split-and-recombine mechanism.

The findings from previous studies suggest that mixing performance in SAR micromixers is significantly influenced by both Reynolds number and channel geometry. Specifically, at lower Reynolds numbers, mixing is primarily diffusion-controlled, whereas at higher Reynolds numbers, secondary flows facilitate enhanced transverse transport, thereby improving mixing efficiency.

Despite the extensive research surrounding SAR micromixers, there remains a paucity of systematic comparisons between circular and rhombus mixing units under consistent numerical conditions. Consequently, this study aims to undertake a controlled numerical comparison of circular and rhombus SAR geometries, alongside a baseline straight configuration, to evaluate the impact of geometry on micromixing performance rigorously.

# 3. Motivation and Objective

## 3.1 Motivation

Efficient mixing is essential in microfluidic systems used in applications such as lab-on-chip devices, biomedical diagnostics, and microreactors. However, mixing in microchannels is inherently difficult because the flow is typically laminar, and mixing occurs primarily through molecular diffusion. Passive micromixers based on the split-and-recombine (SAR) principle provide a practical approach to improve mixing without requiring external energy input.

Previous studies have shown that the mixing performance of SAR micromixers strongly depends on channel geometry. Geometrical modifications can enhance mixing by promoting transverse flow and increasing interfacial area, but they may also lead to increased pressure drop and higher pumping requirements. Therefore, a systematic comparison of different SAR geometries is necessary to identify configurations that provide improved mixing performance while maintaining acceptable hydraulic characteristics.

## 3.2 Objectives

The main objective of the present study is to numerically investigate the effect of geometry on the mixing performance of split-and-recombine microchannel mixers under steady laminar flow conditions.

- The specific objectives of this study are:
- To analyse three microchannel geometries, including a baseline straight channel, a circular SAR channel, and a rhombus SAR channel.
- To evaluate the mixing performance of each geometry over a range of Reynolds numbers using the mixing index.
- To compare the influence of geometry on micromixing behaviour.
- To identify the geometry that provides the most effective mixing performance among the configurations studied.

## 4. Methodology

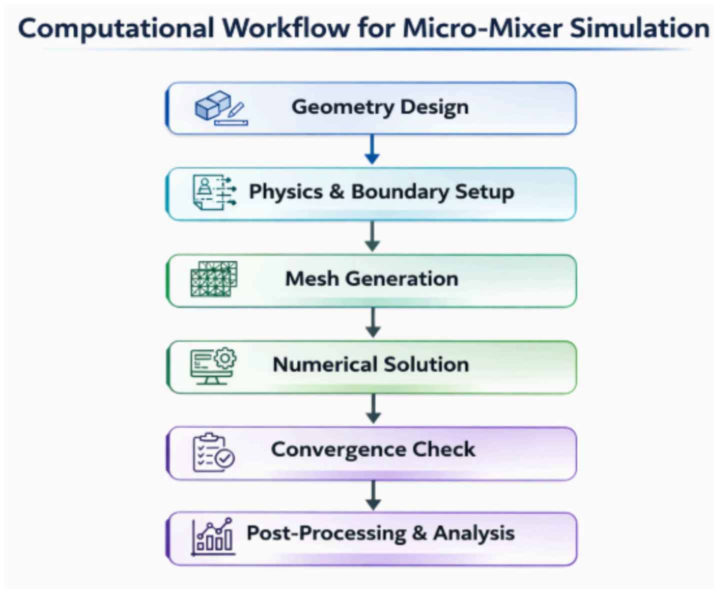



*Fig. 1. Methodology for simulation*

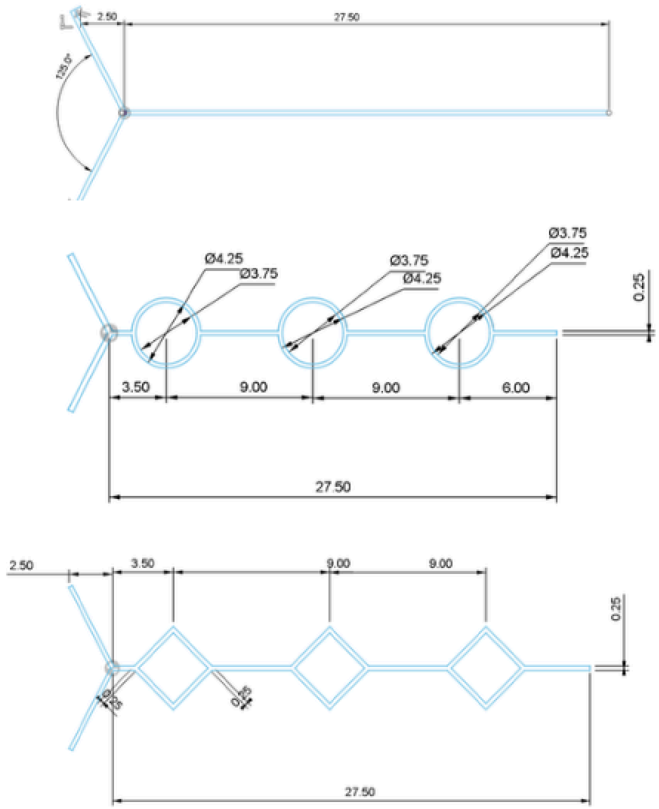
### 4.1 Approach Towards Numerical simulation

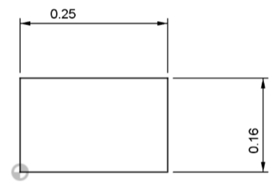
For the purposes of this research , there are six major steps as mentioned in *Fig. 1.* Required during numerical simulation on ANSYS 2025R2.. In brief, In the first step the design is Finalized and is designed using a CAD software , in this case AutoDesk-Fusion360 was used. Then physics and boundary conditions are defined for the domain inlets, outlets, and all walls.

This is followed by Mesh generation and calculation of the solution. Next, the Solution is validated. In the post-processing step data is extracted from the simulations and is further used to visualize the solution results using applications such as MATLAB.

### 4.2 Mixer Design and Meshing

The complete drawings of three-dimensional proposed designs for micromixers are as shown in *Fig. 2.* The total length, width, and depth of micromixers are given by 30mm, 250µm, and 160µm respectively. All the proposed design geometries formed by expanding Y-shaped baseline straight mixer. The shape and size of all micro-mixers were kept the same except the mixing units which are unique for each micro-mixer.




*Cross-section of the channel*

*Fig. 2. Micromixer designs (all dimensions in mm)*

Meshing plays a crucial role in Numerical simulation, Coarse Meshes lead to a faster solution but compensate on accuracy , finer meshes are more accurate but require more time to simulate Therefore, an optimal mesh is required for simulation so that guarantee enough accuracy with low possible computational time.

Hence a mesh element size of 5e-6 or 5µm has been used for this numerical simulation.

## 4.3 Governing equations and Boundary conditions

In this study for evaluation of mixing performance, all proposed designs were simulated by ANSYS 2025R2. We used diluted species transport modules to perform the mixing simulations for three-dimensional models. The governing equations for these physical modes are as presented in Eqs. (1)-(3).

Steady equation/ Continuity equation:

$$\rho \nabla \cdot V = 0 \qquad (1)$$

Navier–Stokes equation:

$$\rho (V \cdot \nabla) V = \nabla \cdot [-P I + \mu (\nabla V + (\nabla V)^T)] + F \qquad (2)$$

Convection–Diffusion equation:

$$\nabla \cdot (-D \nabla C_i) + V \cdot \nabla C_i = R \qquad (3)$$

Boundary Conditions:

$$V = V_0 \quad \text{at inlets} \qquad (4)$$

$$V = 0 \quad \text{at walls} \qquad (5)$$

$$P = 0 \quad \text{at outlet} \qquad (6)$$

where $V$ = fluid velocity ($m\ s^{-1}$), $\mu$ = dynamic viscosity ($Pa{\cdot}s$), $\rho$ = density ($kg\ m^{-3}$),

$P$ = fluid pressure (Pa), $C_i$ = concentration ($mol\ m^{-3}$),

$T$ = viscous stress tensor, $I$ = identity matrix,

$F$ = body force ($N\ m^{-3}$), $R$ = source term and $D$ = diffusion coefficient ($m^2\ s^{-1}$).

Dynamic viscosity, density and diffusion coefficient values used in the simulation setup were 0.001 $Pa{\cdot}s$, $10^3$ $kg\ m^{-3}$ and $1 \times 10^{9}$ $m^2\ s^{-1}$ respectively.

The boundary conditions used at the inlets of the micro-mixer were velocity and the pressure at the outlet. No-slip conditions were imposed at all the walls of the micromixer except inlets and outlet The concentration values at inlet 1 and inlet 2 are given by 0 and 1 $mol\ m^{-3}$ respectively.

The mixing index (M.I) and its formula is given by:

$$M.I = 1 - \sqrt{\sigma^2 / \sigma^2_{max}} \quad (7)$$

For calculating the mixing index from Eq. (7) to compare different mixer designs, the variance of the mass fraction on a cut plane perpendicular to flow at the outlet is given by:

$$\sigma = \sqrt{\left[ (1/n) \Sigma (ai - b)^2 \right]} \quad (8)$$

where n is the number of sample points on the cut plane perpendicular to the flow, ai is the mass fraction value at the sampling point in the cross-sectional plane,b is the optimal mixing value and it is taken as 0.5 for optimal mixing.

As accuracy is very important in evaluating mixing performance, 400 sample points are taken from the cut plane.

For calculating the mixing index, the maximum variance at the inlet of the mixer is calculated by:

$$\sigma_{max} = \sqrt{\left[ b(1 - b) \right]} \quad (9)$$

The higher the value of M.I, the better is the mixing efficiency.

The mixing index range is from 0 to 1. M.I zero represents unmixed situations while 1 represents the fully mixed.

Reynolds number has an important influence on fluid flow and plays an important role in fluid flow problems.It is given by:

$$Re = (\rho v l) / \mu \quad (10)$$

where $\rho$ is the density, v is velocity and l is the hydrodynamic or characteristic length.For the rectangular cross-sectional area of the mixer it is given by:

$$l = (2xy) / (x + y) \quad (11)$$

where x and y are the width and depth of the channel.

The *Table.1.* represents the values for velocity and Hydraulic diameters which are calculated based on the Reynolds number Eq. (10) in the range of $1 \leq Re \leq 3$ with a step size or interval of 0.5 i.e a total of 5 datapoints .

| REYNOLDS NUMBER | HEIGHT | WIDTH | HYDRAULIC DIAMETER | VELOCITY [M/S] |
|---|---|---|---|---|
| 1 | 160 | 250 | 195.122 | 0.005125 |
| 1.5 | 160 | 250 | 195.122 | 0.0076875 |
| 2 | 160 | 250 | 195.122 | 0.01025 |
| **2.5** | **160** | **250** | **195.122** | **0.0128125** |
| 3 | 160 | 250 | 195.122 | 0.015375 |

*Table.1. Reynolds number, inlet velocity, Hydraulic diameter*

## 5. Results and Discussion

In this research we used two types of comparative studies, one of them was that we compared the mixing performance of the three selected geometries at the above given Reynolds numbers by calculating the Mixing Index at the outlet plane given in *subsection 5.1 and Fig.4*.

The other one was in which we selected 2 cut planes in each microchannel at X1=7.5mm and X2=15.5mm from the Y-junction point and analyzed the progressive variation of mixing along each channel at a Re=2.5 as shown in *Fig.3.*

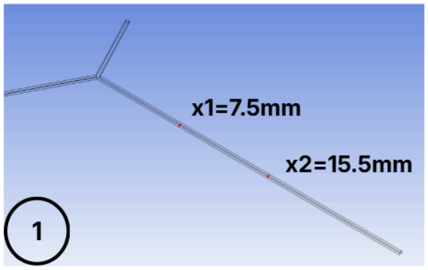
At X1= 7.5mm cut plane only one SAR mixing unit has been crossed as for Geometry 2 and Geometry 3. This is evident from the concentration contours at that cut plane at which the minimum and maximum mass fractions are very similar to those of Geometry 1.

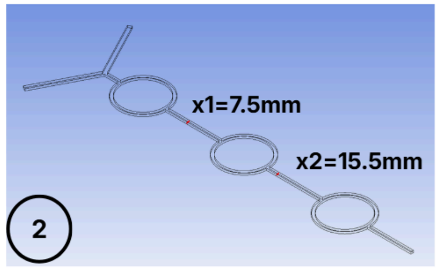
It should also be noted that the range of mass fractions at X1 is very high for all three cases , implying that very minimal or even negligible mixing has taken place. But the mass fraction range in rhombus is slightly less than the other two.

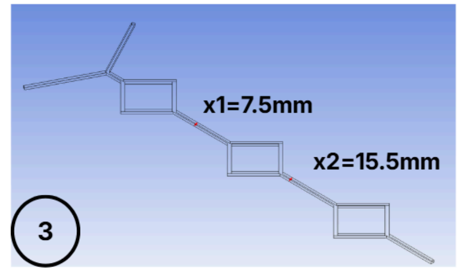
It is observed that the generation of chaotic advection until this point is very limited, as there has only been the effect of one SAR unit in both cases , therefore mixing can be assumed to be dominated by molecular diffusion.

Moving onto the cut plane at X2=15.5mm we see the considerable decrease in the mass fraction ranges of both Geometry 2 and 3 as compared to their prior cases at x1=7.5mm.This reduction indicates that there is an increase in mixing and is the effect of chaotic advection in both cases that is Geometry 2 and 3 , but it is still not as more as we want it to be it.

It is important to note here at the X2 plane that there is a clear difference between the mixing performance of Geometry 2 and 3 compared to Geometry 1 evident from *Fig.3*.

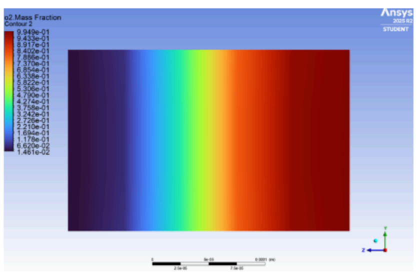


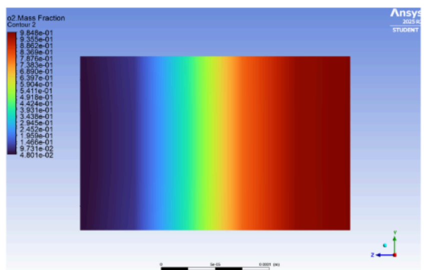

- **CONCENTRATION GRAIDENTS (Re=2.5)**

**x1=7.5mm**



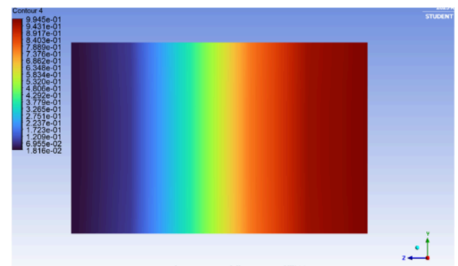
**MIN. MASS FRACTION= 0.014**

**MAX. MASS FRACTION= 0.995**



**MIN. MASS FRACTION= 0.048**

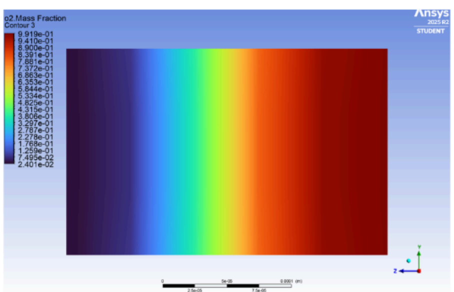
**MAX. MASS FRACTION = 0.985**



**MIN. MASS FRACTION= 0.018**

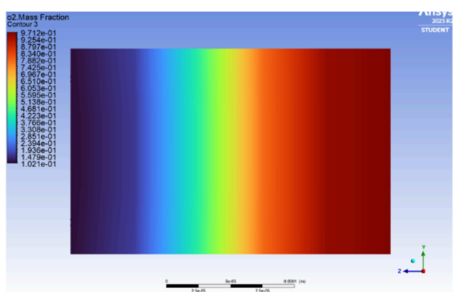
**MAX. MASS FRACTION= 0.993**

**x2=15.5mm**



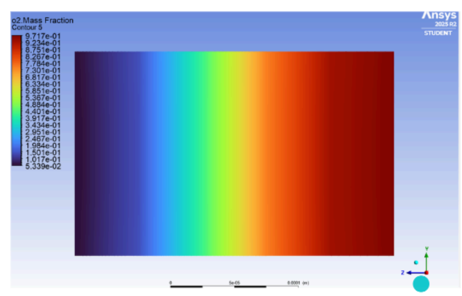
**MIN. MASS FRACTION= 0.024**

**MAX. MASS FRACTION = 0.991**



**MIN. MASS FRACTION=0.103**

**MAX. MASS FRACTION=0.971**



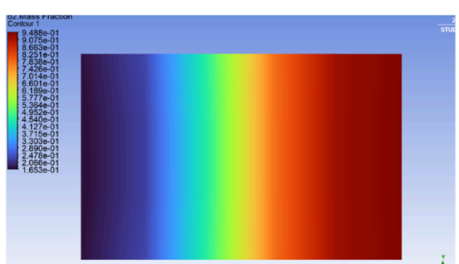
**MIN. MASS FRACTION= 0.097**

**MAX. MASS FRACTION= 0.951**

**x3=outlet**

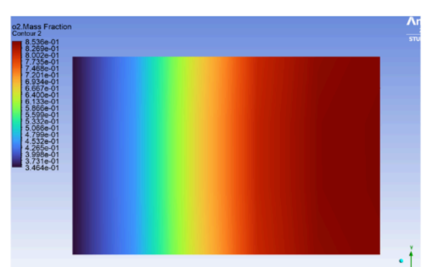
**MIN. MASS FRACTION= 0.037**

**MAX. MASS FRACTION= 0.987**



**MIN. MASS FRACTION= 0.165**

**MAX. MASS FRACTION= 0.948**



**MIN. MASS FRACTION= 0.346**

**MAX. MASS FRACTION = 0.854**

*Fig.3. Comparison of mixing at 7.5mm and 15.5 mm cut plane of each geometry*

While Geometry 1 continues to exhibit limited mixing, the other two geometries demonstrate enhanced mixing..

Nevertheless, at this stage it is still not entirely clear which of Geometry 2 or Geometry 3 generates stronger chaotic advection leading to superior mixing performance.

At X3=27.50mm or at outlet , on comparison of mass fraction ranges , it can be observed that , with the combined effect of three SAR Mixing units in the cases of Geometry 2 and 3 , the mixing performance has considerably risen as compared to at X2 plane in each of the cases.

This signifies the role of generation of chaotic advection of these SAR units in mixing.

It is to be observed that the mixing performance of Geometry 3 can now be distinctly differentiated with other two owing to its high mixing performance at the outlet, implying the superior performance of the Rhombus SAR unit over the Circular SAR unit.

We also see the effect chaotic advection puts on mixing by comparing Geometry 2 and Geometry 3 to the baseline Geometry 1, whose mass fraction remains constant throughout the channel, unlike other two geometries.

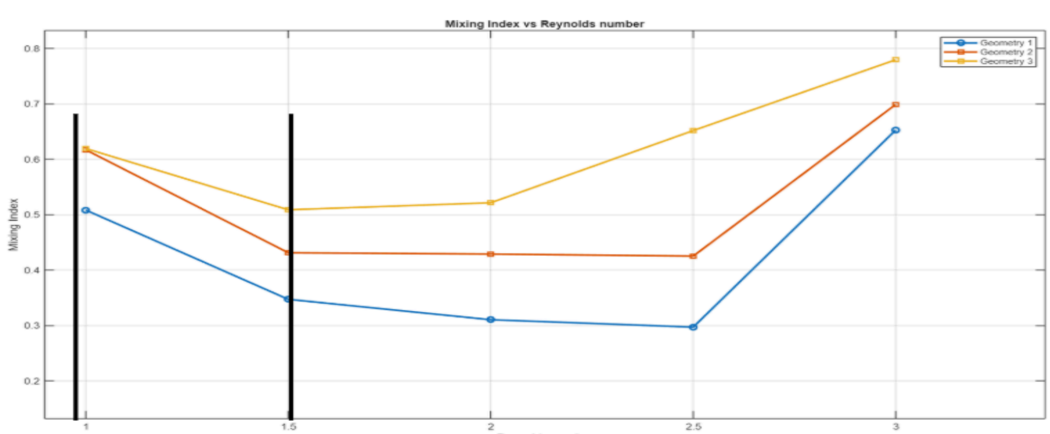
Hence proving the ability of SAR units to enhance mixing by introducing transverse flow and generation of chaotic advection.

## 5.1 Comparative Analysis :

Variation of Mixing Index of each geometry with Reynolds number of the flow.

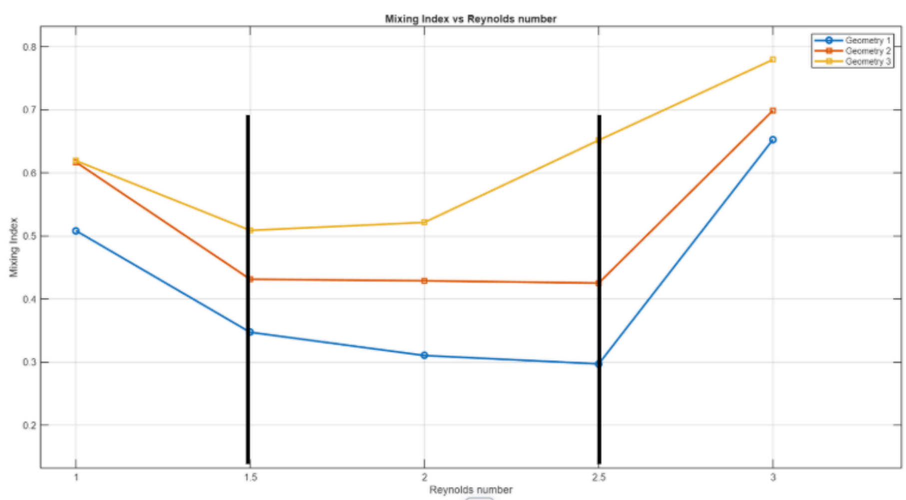
<u>At Low Reynolds Number (Re = 1 to 1.5)</u>

- Flow is strongly laminar and diffusion-dominated.
- Fluid layers move smoothly.
- Mixing happens mainly due to molecular diffusion.
- Since velocity is low, residence time is high, so it has better mixing.
- So the mixing index starts relatively high



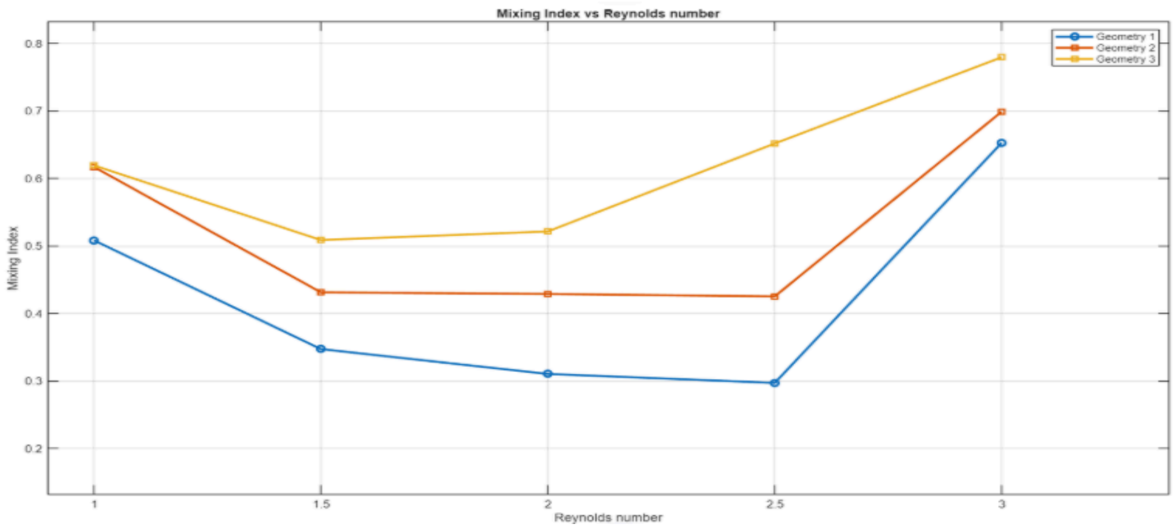
As Reynolds Number Increases (Re = 1.5 to 2.5)

- Flow becomes more convection-dominated but still laminar.
- Velocity increases.
- Residence time decreases.
- Molecular Diffusion has less time to act.
- But the flow is still not strong enough to create vortices or chaotic advection.
- So Mixing becomes worse
- Mixing index decreases
- This region often shows minimum mixing efficiency.




At Higher Reynolds Number (Re = 3)

- Now inertial effects become significant and advection dominates molecular diffusion
- Even if the flow is technically laminar, you may get:
    a. Flow recirculation
    b. Chaotic advection
- These effects
    a. Enhance transverse transport
    b. Promote convective mixing
    c. So the mixing index increases sharply
    d. Transverse flow.




From the comparative analysis of the three geometries, it is evident that the introduction of SAR (Split and Recombine) mixing units significantly enhances mixing performance in microchannels operating at low Reynolds numbers.Among the three geometries, **Geometry 3 (Rhombus SAR unit)** exhibits the highest mixing index at the outlet, indicating superior chaotic advection and transverse flow generation compared to Geometry 2 (Circular SAR unit) and the baseline Geometry 1.

## 6. Conclusion

A comparative numerical investigation was conducted to assess the performance of split-and-recombine (SAR) microchannel mixers across three distinct geometries: a straight baseline channel, a circular SAR configuration, and a rhombus SAR configuration. The simulations were executed under steady-state laminar flow conditions over an array of Reynolds numbers, utilizing the coupled Navier–Stokes and convection–diffusion equations. The mixing performance was quantified using the mixing index.

The findings indicate that mixing efficiency is markedly influenced by channel geometry. The straight microchannel exhibited solely diffusion-dominated transport, resulting in the lowest mixing performance characterized by pronounced concentration gradients at the outlet. Conversely, the SAR geometries significantly enhanced mixing through iterative split-and-recombine operations, thereby facilitating transverse mass transport.

Notably, the rhombus SAR configuration achieved the highest mixing index within the investigated Reynolds number range, attributable to intensified transverse transport and chaotic advection induced by repeated splitting and recombination of fluid streams. The circular SAR geometry similarly demonstrated improved performance compared to the straight channel, due to the generation of secondary flow structures resulting from channel curvature.

The relationship between the mixing index and Reynolds number showed non-monotonic behaviour, underscoring the complex interplay between molecular diffusion and inertial effects on mixing dynamics. Overall, this study provides crucial insights into the geometrical influences on micromixing behaviour, informing the design of more efficient passive micromixers.

# 7. Future Scope

The present study establishes a numerical baseline for evaluating the mixing performance of SAR microchannel mixers. Several extensions can further improve the understanding and performance of SAR micromixers.

Future work may include evaluating hydraulic performance by quantifying pressure drop and defining a performance criterion that balances mixing improvement with pumping power requirements. A wider Reynolds number range may also be investigated, including very low Reynolds numbers where mixing is purely diffusion-dominated and higher Reynolds numbers where inertial effects become more significant.

Further studies may focus on detailed analysis of secondary flow structures through vorticity and Dean number calculations to better understand the mechanisms responsible for mixing enhancement. Parametric studies varying geometric parameters, such as rhombus angle, curvature radius, number of SAR units, and channel aspect ratio, may also be conducted to identify optimised designs.

Additional extensions may include transient flow simulations, non-Newtonian fluid behaviour, and experimental validation through microfabrication and flow visualisation. Advanced optimisation techniques, such as machine-learning-based geometric optimisation, may also be explored to develop high-performance micromixer designs.

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